Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/01/2011
PI Name:	Poirier, David Sc.D.		
Project Title:	Research for ESA Space Experiments on Directional Solidification of Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/15/2008	**End Date:**	05/14/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	2
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Expedition 19/20 NOTE: End date extended to 5/14/2013 (from 5/14/2011) in Dec 2010 per NSSC information (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Erdmann, Robert (University of Arizona) Tewari, Surendra (Cleveland State University)		
Grant/Contract No.:	NNX08AN49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This is a program to conduct experiments in the International Space Station (ISS) to study the effect of microgravity on directionally solidified alloy microstructures and segregation. At Cleveland State University (CSU) terrestrial directional solidification experiments on Al-7%Si alloys have been done using different growth conditions to effect variations in dendritic metrics. Numerical modeling to simulate the transport phenomena and solidification of the planned experiments is currently underway at the University of Arizona (UA). Based on previous research, it is clear that thermosolutal convection masks the effects of diffusional transport during solidification and also leads to macro-segregates known as freckles. We are convinced, therefore, that with terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, it would be possible to effect directional solidification with no thermosolutal convection. Our hypotheses are the convection and attendant macrosegregation will be eliminated and the dendritic microstructure will be greatly altered in microgravity. Availability of microgravity provides an opportunity to obtain experimental data, where thermosolutal convection is negligible. With this grant, growth conditions for such microgravity experiments will be defined by combining mathematical modeling and simulations with the results of terrestrial experiments. Thermosolutal convection will be significantly reduced in microgravity, and thus we can compare segregation and microstructure in specimens terrestrially solidified and those solidified in microgravity. The technological significance of the research is that it will provide scientific underpinning to our understanding and control of how single-crystal castings solidify. Single-crystal castings are critical components in high-temperature gas-turbine engines that are used in high-speed aircraft and land-based power generators. Currently we are doing research on the effect of convection on the microstructure and macrosegregation of directionally solidified castings. We are part of MICAST (Microstructure Formation in Casting of Technical Alloys) led by ESA-researchers in the German Aerospace Center in Germany. One sample was processed in the ISS in August, 2010 and returned to Earth. Its microstructure is under study at CSU and MSFC. A second cartridge was processed on the ISS in Feb. 2011, returned to Earth, and as of March, 2011, is now at MSFC ready to be opened.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is relevant to a process known as directional solidification, which is used for producing critical castings that are used in gas turbine engines. Gas turbine engines power aircraft (i.e., often called "jet engines") and are also used to generate electric power. These castings are made of vacuum melted alloys that are capable of withstanding the demanding conditions at high temperatures and consequently the metallurgical structure of the alloys must be of the highest quality. The research will lead to an improved understanding of how microstructures and segregation form in directionally solidified castings; hence the science of producing such castings will be enhanced as a result of the research. In addition, the research-findings will benefit many other types of castings, including billets and ingots, of alloys (i.e., steel, aluminum alloys, etc.) that are important in the "manufacturing chain."
Task Progress:	One sample was processed in the ISS in August, 2010 and returned to Earth. Its microstructure is under study at CSU and MSFC. A second cartridge was processed on the ISS in Feb. 2011, returned to Earth, and as of March, 2011, is now at MSFC ready to be opened.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Erdmann RG, Hendrick AG, Goodman MR. "Properties of stochastic permeability." Transactions of The Indian Institute of Metals 2009 Aug-Oct; 62(4-5):261-4. http://dx.doi.org/10.1007/s12666-009-0038-5 , Oct-2009
Articles in Peer-reviewed Journals	Erdmann RG, Poirier DR, Hendrick AG. "Permeability in the mushy zone." Materials Science Forum. 2010 May;649:399-408. http://dx.doi.org/10.4028/www.scientific.net/MSF.649.399 , May-2010
Dissertations and Theses	Goodman MR. "Properties of Stochastic Flow and Permeability of Random Porous Media." Thesis, The University of Arizona, Tucson, April 2010. , Apr-2010